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"Digital Gold" and geopolitics

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A B S T R A C T

There is a growing empirical literature on Bitcoin and gold safe haven properties with respect to financial risks and macroeconomic news but very scarce literature regarding geopolitical risks. This paper provides a fresh insight into the Bitcoin safe haven status, in comparison to gold. We, first, propose a geopolitical risk composite indicator based on various sources of geopolitical risks. A Principal Component Analysis is conducted to group the information on these indicators. Second, a dynamic Markov-switching copula model (which accommodates a dynamic link between the developed geopolitical risk index and Bitcoin and gold price dynamics within low and high risk regimes) is used. We show that both Bitcoin and gold respond positively to the composite geopolitical risk indicator when risk is high. This underscores that both Bitcoin and gold have the ability to act as safe havens for assets whose valuations plummet during times of violent geopolitical conflicts. But such properties seem to be conditional upon different categories of geopolitical risks.

1. Introduction

Geopolitical risks (GPRs) have long been viewed as a potential driving force of asset price dynamics. Accordingly, Caldara and Iacovielloz (2018) argued that geopolitical uncertainty could have significant impacts on asset price changes. Even though the empirical literature on the nexus between various uncertainty indicators and assets is rising substantially (inter alia: Balcilar et al., 2016; Beckmann et al., 2017; Bouoiyour et al., 2018, among others), the impacts of geopolitical risks on financial and macroeconomic variables has not largely been explored. The majority of empirical researches have focused on the role of geopolitical events in determining movements of financial market returns and volatility (see, for instance, Eldor and Melnick, 2004; Arin et al., 2008; Nguyen and Enomoto, 2009; Drakos, 2004; Karolyi and Martell, 2010; Kollias et al., 2010, 2013; Chesney et al., 2011, etc.).

The analysis of geopolitical risks fills a gap in the growing empirical literature on safe haven assets and gold as most studies focus on gold's safe haven status against financial variables such as changes in stock market indices (Baur and McDermott, 2010), bond market indices (Baur and Lucey, 2010), exchange rates (Reboredo, 2013a) and oil prices (Reboredo, 2013b). Notably, the majority of studies examine gold and there are only a few studies that compare the safe haven abilities of Bitcoin and gold (Al-Khazali et al., 2018; Selmi et al., 2018; Bouoiyour et al., 2019a,b; Będowska-Sójka and Kliber, 2021). Bitcoin has shown great resilience during periods of turmoil, highlighting its potential hedging and safe haven abilities against global uncertainty. Bouri et al. (2017a) claim that the global uncertainty surrounding the 2008 global financial collapse eased the emergence of Bitcoin and strengthened its popularity as both a

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financial asset and an alternative currency to conventional economies. Luther and Salter (2017) indicated that attention towards Bitcoin rose remarkably following the announcement that Cyprus would accept a bailout on March 16, 2013. Bitcoin has also been reported in countries such as Greece, whose banks are troubled. Other recent studies that have analyzed the role of Bitcoin as a hedge against various assets (Popper, 2015), against commodities (Dyhrberg, 2016) and against financial stress (Bouri et al., 2020a). This has logically led to a comparison between Bitcoin and gold, as gold is largely regarded, in theory, as a hedge and safe haven to protect against similar risks (for example, Al-Khazali et al., 2018; Będowska-Sójka and Kliber, 2021). In contrast, as far as the impact of geopolitical risks on Bitcoin and gold price dynamics is concerned, to the best of our knowledge, there has been a limited literature that emphasize the importance of geopolitical events in determining gold and Bitcoin price dynamics (for instance, Baur and Smales, 2018; Bouri et al., 2020b; Mamun et al., 2020; Su et al., 2020).

Our study goes beyond earlier work by proposing a new geopolitical risk composite indicator (GRCI) aimed at synthesizing the impacts of seven risk factors (namely US-China relations, global trade tensions, Gulf tensions, European fragmentation, North Korea conflict, Russia-NATO tensions and major terror attacks) within one aggregate measure of risk. This index is a broad measure of global geopolitical risk, as it incorporates not only terrorist attacks but also other forms of geopolitical tensions like military threats, European fragmentation, Gulf tensions and global trade war. Such an index enables us to effectively detect geopolitical risks of several forms all over the world and helps thus to go beyond the impact of specific events in a specific country at a specific point in time. Moreover, the newly geopolitical risk index considers both the direct effect of geopolitical events by accounting for military-related tensions involving large regions of the world and the nuclear tensions, and the effect of 'pure' geopolitical risks while incorporating the geopolitical events (not just risks) which can be expected to prompt a heightened geopolitical uncertainty including terrorist acts. Our analysis is also complementary to a large stand of literature that examine the economic effects of wars, terror attacks, and other forms of collective violence including Tavares (2004); Glick and Taylor (2010), and Moretti et al. (2014).

Importantly, this paper goes beyond evaluating the reactions of gold and Bitcoin prices to some international events. Rather, it assesses gold and Bitcoin safe haven statuses against the core geopolitical risk effects. It must be stressed at this stage that there has been very little research reported on the development of an aggregate measure of the uncertainty using a variety of indicators (Haddow et al., 2013; Charles et al., 2017), but only one study focuses on the effect of a composite uncertainty index synthesizing the impact of a composite indicator summarizing different indexes (i.e., economic, macroeconomic, microeconomic, monetary policy, financial and political uncertainties) on asset returns (Bouoiyour et al., 2018).

Furthermore, it is widely known that asset price dynamics are greatly challenging as they inherently include stochastic and nonlinear components. However, several studies on the behavior of Bitcoin and gold prices in uncertain times have ignored the shifts of this dependence. The present research tries to fill this gap by using a new copula-based approch able to shed new light on the dynamic dependence between Bitcoin returns and geopolitical risks under low and high risk periods, in comparison to gold. In particular, we employ alternative modelling approaches based on some recent studies that use copula classes and Markov-switching models (for example, Balcilar et al., 2016). The copula models ease separate modeling of the marginal distributions and the dependence and therefore, a variety of dependence structures can be detected with more flexibility and parsimony. This technique enables us to reproduce extreme return clustering and asymmetry by allowing for two time-varying dependence regimes, low or normal and high or crash, at the center and tails of the bivariate distribution. The copula-based approach used throughout this study generalizes existing Markov-switching models by allowing for distinct mean reversion in dependence within each state. This method is appealing as the behaviour of time series often appear to go through different phases. For investors wanting to use Bitcoin and gold safe haven characteristics to protect against heightened geopolitical tensions and sudden geopolitical events, it seems very useful to discover whether the geopolitical impact on Bitcoin and gold prices is consistent across all states, or concentrated in specific times of geopolitical turmoil.

We provide robust evidence of a positive response of Bitcoin and gold prices to the different geopolitical risk indicators under high risk scenario, though with varying degrees. This highlights that both Bitcoin and gold would serve as a buffer against exposure to unforeseen geopolitical events. But these safe haven features are likely to be sensitive to various risk levels and the different sources of geopolitical uncertainty.

The remainder of the study is organized as follows: Section 2 reviews the related literature. Section 3 describes the empirical strategy and presents the data and the descriptive statistics. Section 4 reports and discusses the empirical estimation results drawn using a range of geopolitical risk indicators. Section 5 concludes and suggests new path of future research.

2. Literature review

The investigation of the financial capabilities of an asset usually takes into account the liquidity, the reactivity to the variance of other assets, the arbitrage and diversification opportunities and the hedging and safe haven abilities. Such assessment will provide an accurate view of what place the investigated asset has comparably to other assets. Previous research has examined the liquidity and means of exchange of Bitcoin, the diversification possibilities and the arbitrage possibilities . Over the last two years, Bitcoin has repeatedly been compared to gold as it has much gold-like properties (Al-Khazali et al., 2018; Selmi et al., 2018; Bouri et al., 2020b; Będowska-Sójka and Kliber, 2021) both assets have excessive price volatility and the total supply is finite. The non-existence of counterparty risk for gold is, however, not matched by Bitcoin's peer-to-peer payment network (Baur et al., 2015). As gold has well-known hedging and safe haven abilities against stocks, bonds and the American dollar, Bitcoin might exhibit similar features. In fact, the increased Bitcoin prices in combination with heightened economic and geopolitical uncertainty have attracted the attention of investors, risk managers and the financial media.

Even though the literature has largely documented how precious metals (with large extent gold) can act as a safe haven during financial turmoil (Baur and Lucey, 2010), the literature dealing with the role that may be played by Bitcoin as a hedging investment is scare. For instance, Dyhberg (2016) examines the hedging capabilities of Bitcoin. The author suggests that Bitcoin possess hedging characteristics as gold and can be incorporated in a portfolio to mitigate the adverse effects of risks. Baur et al. (2015) assess the statistical properties of Bitcoin and document that it seems uncorrelated with traditional asset classes such as stocks, bonds and commodities in normal times and in periods of financial turmoil. The author claims that the observed dissimilarity in the properties of Bitcoin and traditional asset classes may provide better diversification benefits. In the same context, Bouri et al. (2017 b) utilize a dynamic conditional correlation model to address whether Bitcoin can serve as a hedge and a safe haven for the major world stock indices, bonds, oil, the general commodity index and the U.S. dollar index. They show that Bitcoin can act as a poor hedge. They also find that Bitcoin can also serve as a strong safe haven but solely against extreme down movements in Asian stocks. But they find these hedging and safe haven capabilities to vary across time-horizons. Besides, Bouoiyour and Selmi (2019) seek to address whether the uncertainty surrounding the the debt crisis exert a significant effect on the interest in Bitcoin as an alternative to conventional and state-sanctioned banking system. They provide strong evidence of a great resilience of Bitcoin during periods of turmoil, highlighting its potential safe haven ability against uncertainty.

Moreover, Bouri et al. (2017c) empirically gauge the linkage between Bitcoin and commodities by evaluating the role of Bitcoin as a diversifier, hedge, or safe haven against daily movements in energy commodities and non-energy commodities. They find that Bitcoin can acts as effective hedge and a safe haven against movements in energy commodity indices. However, Bitcoin cannot serve as a hedge and safe haven against non-energy commodities. By considering the December 2013 Bitcoin price crash, they show that Bitcoin keeps its status as a hedge and a safe haven for energy commodities, while it loses this ability in the post-crash period. They also demonstrate heterogeneity in the dynamic correlations between the extreme downward and extreme upward movements.

A limited strand of literature has attempted to compare the hedge, safe haven and diversification properties of Bitcoin and gold in regard to various sources of uncertainty and at various time-horizons. Bouoiyour et al. (2019a) carry out a new approach to modeling dynamic correlations between Bitcoin and gold returns. They show a positive and strong correlation between these markets in times of economic turmoil, but after the normalization of the situation or under bullish market regimes, this relationship loses its relevance. They attribute the positive correlation between Bitcoin and gold to the fact that both Bitcoin and gold represent investments that people seek because they are safe and risk minimal. Al-Khazali et al. (2018) compare the features of Bitcoin and gold in regard to unforeseen macroconomic events. To do so, they assessed the asymmetric responses of their returns and volatility to positive and negative macroeconomic news surprises originated from the US, Canada, the Euro Area, UK, and Japan. They show that the effect of both negative and positive unusual macroeconomic innovations is more pronounced for gold than for Bitcoin. Besides, Bouri et al. (2020a) assess the safe-haven properties of Bitcoin, gold, and the commodity index against world, developed, emerging, USA, and Chinese stock market indices for different time scales. They show that all these assets have time-varying safe haven characteristics, with superiority of Bitcoin. Recently, the heightened uncertainty surrounding the Covid-19 and the resulted financial turbulence has yielded to increased attention towards the hedging, safe haven and diversification features of different financial assets, some assessing cryptocurrency implications (Alfaro et al., 2020; Conlon et al., 2020; Conlon and McGee, 2020; Corbet et al., 2021; Goodell and Goutte, 2021a, b). Conlon and McGee (2020) provide an accurate evaluation of the safe haven characteristics of Bitcoin for a traditional asset during times of acute equity price collapses. Besides, Conlon et al. (2020a) look at the investment benefits (in particular, the safe haven properties) of Bitcoin and other major cryptocurrencies for different international equity markets in times of high uncertainty over the COVID-19 pandemic. They found that only investors in the Chinese CSI 300 index benefited from a limited allocations to Bitcoin. Interestingly, Goodell and Gouttle (2021a) have tried to assess how the role of Bitcoin in hedging other assets has changed in times of increasing anxiety over the COVID-19 pandemic. Addressing how Bitcoin prices co-move with the intensity of the COVID 19 crisis under different time-horizons (by means of wavelets) has contributed significantly to the existing literature on the hedging, safe haven and diversification opportunities of Bitcoin in times of distress due to unusual events. Moreover and using different econometric tools, Goodell and Gouttle (2021b) tested if cryptocurrencies offer diversification benefits for international equity markets. They showed that co-movements between cryptocurrencies and equity indices increased as COVID-19 progressed. Nevertheless, these linkages are likely to be relatively modest, underscoring that cryptocurrencies (in particular, Bitcoin, Ethereum, Litecoin and Tether) are not highly suitable for diversifying purposes during both normal times or downturns. Moreover, Będowska-Sójka and Kliber (2021) explore the safe-haven features of gold, Bitcoin and Ether in regard to heightened uncertainty surrounding several events. More particularly, they compare the responses of these assets to various downturns on the financial markets, including the 2015/2016 Chinese economic downturn and the recent COVID-19 pandemic. They find the superiority of gold especially in times of COVID anxiety.

However, as far as the reactions of Bitcoin and/or gold to geopolitical risks is concerned, to the best of our knowledge, few studies have tried to test the safe haven abilities of Bitcoin and gold with respect geopolitical uncertainty. For instance, Baur and Smales (2018) document that gold responds positively to global geopolitical risks and geopolitical threats but no significant reaction to realized geopolitical acts. Al-Yahyaee et al. (2019) claim that geopolitical risks exert a significant influence on the co-movements among the volatility index (VIX) and the Bitcoin prices. Bouoiyour et al. (2019b)'s study underscore the ability of Bitcoin to serve as a safe haven in times of heightened uncertainty surrounding the 2016 U.S. presidential elections results. Using a truly noise-assisted data analysis method, termed as the Ensemble Empirical Mode Decomposition-based approach, they find that Bitcoin's safe-haven property for the U.S. stock market is time-varying and that this digital cryptocurrency serves as a weak safe haven in the short term, and as a hedge in the medium- and the long-run. They also document that precious metals lost their safe haven properties over time as the correlation between gold/silver and the U.S. stock prices increases from the short- to long-term horizons. Bouri and Gupta (2019) argue that Bitcoin may act as a hedge against uncertainty which is caused, even partially, by the global geopolitical risks. More recently, Mamun et al. (2020) show that the impact of geopolitical uncertainty on Bitcoin price dynamics are more pronounced during unfavorable economic circumstances. Su et al. (2020) conducted sub-sample rolling-window Granger causality tests to explore, from a

Table 1
Descriptive statistics of return series.

	Mean	Median	Std. Dev.	Skewness	Kurtosis	Jarque-Bera	p-value
BTR	0.0314	0.0186	6.1257	−0.0500	3.6500	11.2636	0.0014
GR	0.0249	0.0168	2.1345	−0.2351	6.0366	78.682	0.0056
UCR	0.0216	0.0186	4.1134	−0.0500	3.6500	11.2636	0.0014
GTD	0.0265	0.0257	0.0393	−0.086	3.6753	11.4933	0.0068
GT	0.0345	0.0512	2.0341	−0.0371	4.2356	13.892	0.0039
EF	0.0148	0.0127	2.1956	−0.1950	3.6266	24.778	0.0097
NKC	0.0345	0.0225	0.0991	−0.4516	4.1567	21.489	0.0000
RNT	0.0168	0.0808	3.3458	−0. 1764	4. 5175	15.855	0.0000
TA	0.0411	0.0321	0.0156	−0.0672	4.1024	19.052	0.0234

Notes: Std. Dev. symbolizes the Standard Deviation; p-value corresponds to the test of normality based on the Jarque-Bera test; *BTR*: Bitcoin returns; *GR*: Gold returns, *GTD*: the global trade index; *UCR*: the US-China relations risks index, *EF*: the uncertainty related to the European fragmentation; *GT*: the uncertainty over Gulf tensions; *NKC*: the uncertainty over the North Korea conflict; *RNT*: The uncertainty over Russia-NATO tensions; *TA*: the uncertainty surrounding major terror attacks.

new perspective, the causal relationship GPRs and Bitcoin prices. They find that there is a positive causal link that runs from Bitcoin price dynamics to GPRs, highlighting that the Bitcoin market is a leading indicator that can be employed to effectively assess the global geopolitical environment. Interestingly, by investigating the discontinuity in Bitcoin prices (and other major cryptocurrencies) and wide swings in GPRs, Bouri et al. (2020b) results reveal a significant co-jumps for the case of Bitcoin, highlighting that Bitcoin serves as hedge against rising geopolitical risks.

The aforementioned considerations have motivated us to compare the hedging and save haven properties of Bitcoin to gold, since the shiny metal is perceived in theory as a hedge and a safe haven to protect other assets against similar risks with an application to geopolitical risks, while differentiating between the dynamic dependencies of extreme downward and extreme upward movements.

3. Data and methodology

3.1. Data and descriptive statistics

Market attention to geopolitical risk is still elevated. This study examines the dynamic relationship between Bitcoin returns (*BTR*) and a developed geopolitical risk composite index (*GRCI*), conditional upon distinct risk scenarios (i.e., low and high regimes), in comparison to gold (*GR*). As for Bitcoin daily prices, they are extracted from Coindesk (www.coindesk.com/price) that computes the average price across leading Bitcoin exchanges, and therefore it detects global Bitcoin prices better than other alternatives. Bitcoin prices are denominated in USD. The data for gold prices, which are measured in USD per ounce, were downloaded from the website of the World Gold Council (https://www.gold.org/goldhub/data/gold-prices). We also use new geopolitical risk indicators proposed by BlackRock Investment Institute (https://www.blackrock.com/corporate/insights/blackrock-investment-institute/interactive-charts/geopolitical-risk-dashboard). These indicators are constructed by identifying particular words associated to geopolitical risk and employ text analysis to calculate the frequency of their appearance in the Thomson Reuters Broker Report, Dow Jones Global Newswire databases and Twitter. The search was categorized as heightened trade tensions including Global trade (*GTD*) and US-China relations risks (*UCR*), and increased risks from the European fragmentation (*EF*) as well as market focus on risks such as Gulf tensions (*GT*), North Korea conflict (*NKC*), Russia-NATO tensions (*RNT*) and major terror attacks (*TA*).Table A1 (Appendix) reports all the data used, their definition, their availability and their sources. To be consistent in our analysis, we consider similar frequencies and time periods for all the considered variables. In particular, we use daily data for the period. To do so, we utilize the first-differenced natural logarithm of We consider daily price data for the Coin Desk Bitcoin Price Index and the gold price ranging from January 3, 2010 to December 26, 2019.

Table 2
The BDS test based on the residuals of the return series.

Dimensions	m = 2	m = 3	m = 4	m = 5	m = 6
BTR	1.9823**	2.9304**	3.1552***	3.0925**	2.9250***
GR	2.1368**	3.1831**	3.1546**	2.3768***	3.1267**
UCR	2.0772*	2.6284***	3.3410***	4.0333***	4.2134***
GTD	2.3451**	2.5410**	3.1962***	3.4182***	3.2456***
GT	8.0844***	9.6261***	10.397***	10.812***	11.324***
EF	1.5965*	1.0347*	1.1904*	1.3315*	1.3112*
NKC	5.7185***	6.0511***	7.4964***	8.7352***	9.8141***
RNT	1.9719**	6.4523**	4.3512**	6.7921***	8.0412***
TA	1.9452*	1.9123*	4.1562**	3.9124***	4.1278***

Notes: *BTR*: Bitcoin returns; *GR*: Gold returns, *GTD*: the global trade index; *UCR*: the US-China relations risks index; *GTD*: the global trade index; *GT*: the uncertainty over Gulf tensions; *EF*: the uncertainty related to the European fragmentation; *NKC*: the uncertainty over the North Korea conflict; *RNT*: The uncertainty over Russia-NATO tensions; *TA*: the uncertainty surrounding major terror attacks; *m* denotes the embedding dimension of the BDS test. ***, ** and *indicate the rejection of the null of the residuals being *iid* at the 1%, 5% and 10 % levels of significance, respectively.

Table 1 provides the descriptive statistics for the daily returns of the variables of Bitcoin and gold returns and changes in the different geopolitical risk indexes under study. We show that the average daily returns of all the time series under study is positive. The mean returns are close to zero for all of the return series and appear small relative to their standard deviations, which would imply that there is no significant trend in the data. The standard deviation values indicate that Bitcoin returns, US-China relations indicator and Russia-NATO tensions index are highly volatile, to a lesser extent the gold returns, and changes in the European fragmentation proxy and Gulf tensions index. The skewness coefficient for all the variables is negative and the kurtosis coefficients is above three, indicating that the probability distributions of the considered return series are skewed and leptokurtic, thereby rejecting normality. All the return series are non-normal as indicated by the Jarque-Bera test.

We, then, test the occurrence of nonlinearities in the considered time series using the BDS test (Brock et al., 1996) of nonlinearity on the residuals recovered from the OLS models. The findings are summarized in Table 2 which show strong evidence of nonlinearity, as they reject the null hypothesis of independent and identical distribution (i.i.d). The results suggest that BTR, GR and all the geopolitical risk indices are non-linearly dependent, which is one of the indications of a chaotic behavior. Such outcome justifies the usefulness of copula with regime switching for examining the nonlinear relationship between geopolitical risks and Bitcoin/gold returns.

3.2. *The dynamic copula with markov-switching*

The use of econometric approaches that acknowledge shifts in the dependence structure between Bitcoin/gold returns and geopolitical risk indicators can be very useful and beneficial to limit systemic risks during stressed market circumstances. In this study, we carry out a dynamic Markov-switching copula model which accommodates a flexible relationship among time series within each regime. This characteristic distinguishes this technique from conventional copulas and standard switching regime models where a function is presumed to govern each regime despite how long the given state prevails. Several empirical studies have linked the low dependence regime with Gaussian copula which supposes zero tail dependence to the high dependence regime with non-Gaussian copula that allows tail dependence. For example, the standard Markov-switching copula initiated by Rodriguez (2007) assumes that the means, variances and correlations shift simultaneously and that each regime is dictated by a different static copula. Likewise, Chollete et al. (2009) and Garcia and Tsafack (2011) propose dependence models which are parameterized by static Gaussian copula in normal conditions and a mixture of static elliptical/Archimedean copulas in a bearish regime (or the regime with high volatility).

Following Fei et al. (2013) and while trying to outline the dynamic copula with Markov-switching framework, let S_t be a state variable that represents the prevailing regime. The joint distribution of X_{1t} and X_{2t} conditional on being in regime s is expressed as follows:

$$\left(X_{1t}, X_{2t} \middle| X_{1,t-1}, X_{2,t-1}; S_t = s\right) \approx C_t^{S_t}\left(\mu_{1t}, \mu_{2t} \middle| \mu_{1,t-1}, \mu_{2,t-2}; \theta_t^{S_t}\right) \tag{1}$$

where $s \in \{H, L\}$ where H is the high dependence regime and L the low dependence regime. The random variable S_t follows a Markov chain of order one distinguished by the transition probability matrix denoted as:

$$\pi = \begin{pmatrix} \pi_{HH} 1 - \pi_{HH} \\ 1 - \pi_{LL}\pi_{LL} \end{pmatrix} \tag{2}$$

where π_{HH} and π_{LL} are the so-called staying probabilities, namely, π_{HH} (π_{LL}) is the probability of being in the high (low) dependence regime at time t conditional on being in the same regime at time t − 1.

Previously, Patton (2006) sets out the foundations for time-varying copulas by demonstrating Sklar's theorem for conditional distributions, and claimes the generic copula dependence parameter θ evolves as a ARMA fashion as follows which permits mean-reversion in dependence based on the forcing variable Γ_t.

$$\theta_t = \wedge(\omega + \varphi\theta_{t-1} + \psi\Gamma_t) \tag{3}$$

The dynamic copula with Markov-switching recently developed consists first to apply a regime-switching ARMA copula where the dependence structure evolves as follows:

$$\theta_t^{S_t} = \wedge\left(\omega^{S_t} + \varphi\theta_{t-1}^{S_{t-1}} + \psi\Gamma_t\right) \tag{4}$$

where Λ (·) is the modified logistic or exponential function.[1]

Thereafter, a regime-switching dependence model is carried out where the time-varying copula function that governs each regime is of DCC type expressed as follows:

$$Q_t^{S_t} = \left(1 - \varphi^{S_t} + \psi^{S_t}\middle|\bar{Q} + \varphi S_t Q_{t-1}^{S_{t-1}} + \psi^{S_t}\varepsilon_{t-1}\right) \tag{5}$$

[1] In the context of elliptical copulas, the dynamic parameter is the conventional correlation measure, θt = ρt, and Λ (y) = $(1 - e^{-y})(1 + e^{-y})^{-1}$ is the modified logistic transformation to ascertain $\rho_t \in (-1, 1)$. Once these parameters are estimated, they can be mapped into time-varying rank-correlation and tail dependence measures. In the SJC copula, the parameters modeled in (2) are directly the upper tail dependence, $\theta_t = \lambda^U{}_t$ (or lower tail dependence $\theta_t = \lambda^L{}_t$), and Λ (y) = $(1 + e^{-y})^{-1}$ is the logistic transformation to ensure $\lambda^U{}_t, \lambda^L{}_t \in (0, 1)$.

Table 3
Estimation of the dynamic copula with Markov-switching: Bitcoin (vs. gold) and geopolitical risks.

	Bitcoin and geopolitical risks indicators						
	BTR-UCR	BTR-GTD	BTR-GT	BTR-EF	BTR-NKC	BTR-RNT	BTR-TA
ω_H	0.1642* (0.0396)	0.1098*** (0.0004)	0.2186 (0.2391)	0.1652** (0.0039)	0.1681** (0.0041)	0.2341 (0.1785)	0.0682*** (0.0000)
ω_L	0.0811* (0.0964)	0.1342 (0.1095)	0.1092 (0.1685)	−0.0245 (0.3461)	0.1342 (0.1689)	0.0982* (0.0310)	0.1123 (0.1672)
φ	0.1467*** (0.0001)	0.1812* (0.0319)	0.1942* (0.0134)	0.1019** (0.0067)	0.1142*** (0.0000)	0.1721** (0.0038)	0.0146*** (0.0000)
ψ	0.1561** (0.0072)	0.1321** (0.0054)	−0.0152 (0.2465)	0.1362* (0.0149)	0.1577* (0.0143)	0.1910*** (0.0000)	0.0311** (0.0072)
π_{HH}	0.2309* (0.0121)	0.6934*** (0.0004)	0.8617*** (0.0000)	0.4971** (0.0038)	0.8123*** (0.0000)	0.7312* (0.0145)	0.7155** (0.0049)
π_{LL}	0.2134* (0.0106)	0.2341 (0.6510)	0.1972** (0.0062)	0.2246*** (0.0000)	0.4921*** (0.0009)	0.2241*** (0.0000)	0.3142*** (0.0000)
	Gold and geopolitical risk indicators						
	GR-UCR	GR-GTD	GR-GT	GR-EF	GR-NKC	GR-RNT	GR-TA
ω_H	0.1345** (0.0061)	0.1176** (0.0093)	0.1415* (0.0481)	0.1378*** (0.0004)	0.1078 (0.1132)	0.1583*** (0.0000)	0.0335* (0.0159)
ω_L	0.0835** (0.0056)	0.0668* (0.0541)	0.0167* (0.0378)	0.0955 (0.1347)	0.1125 (0.2481)	0.1062* (0.0418)	0.0562* (0.0344)
φ	0.1921** (0.0050)	0.1413*** (0.0000)	0.1612*** (0.0004)	0.0414* (0.0302)	0.1612*** (0.0004)	0.1861*** (0.0000)	0.0391** (0.0043)
ψ	0.2141*** (0.0005)	0.1092*** (0.0001)	0.1231* (0.0106)	0.0510*** (0.0002)	0.1456** (0.0011)	0.2235* (0.0142)	0.0715* (0.0293)
π_{HH}	0.8723*** (0.0000)	0.9154* (0.0169)	0.9346** (0.0052)	0.8972* (0.0109)	0.7685*** (0.0002)	0.8922** (0.0019)	0.3127*** (0.0008)
π_{LL}	0.2395** (0.0046)	0.2895*** (0.0013)	0.3745* (0.0357)	0.3613** (0.0054)	0.3249** (0.0041)	0.3045*** (0.0007)	0.8012** (0.0029)

Notes: *BTR*: Bitcoin returns; *GR*: Gold returns, *UCR*: the US-China relations risks index; *GTD*: the global trade index; *GT*: the uncertainty over Gulf tensions; *EF*: the uncertainty related to the European fragmentation; *NKC*: the uncertainty over the North Korea conflict; *RNT*: The uncertainty over Russia-NATO tensions; *TA*: the uncertainty surrounding major terror attacks; $\omega_H(\omega_L)$ correspond to the level of dependence between Bitcoin/gold returns and geopolitical risk proxies in crisis period (low volatility or normal states). φ and ψ refer to the rank correlations; Superscript *H (L)* indicates the high (low) dependence regime. π_{HH} (π_{LL}) is the probability of staying in the high (low) dependence regime. Notes: *, ** and***denote statistical significance at the 10 %, 5 % and 1 % levels, respectively.

with $Q_t^{S_t}$ the auxiliary matrix determining the rank correlation dynamics, $\varphi^{S_t} + \psi^{S_t} \prec 1; \varphi^{S_t} + \psi^{S_t} \in (0, 1)$. Note that the Gumbel copula is used to allow for time-variation in dependence and tail dependence within each regime. This would permit us to capture sharp increases (decreases) in dependence in turbulent (tranquil) states without imposing the restriction of static within-regime dependence.

4. Empirical results

4.1. The dynamic copula with Switching regime results

An asset could serve as a safe haven in uncertain times when it is positively correlated with risk indicators under high risk scenario (Jones and Sackley, 2016; Bouoiyour et al., 2018). Accordingly, we examine the responses of Bitcoin returns to distinct geopolitical risk categories based on Markov switching model synchronizing different episodes of geopolitical risk (i.e., low- and high-risk regimes). For comparison purpose, the same exercise is then conducted for gold returns.

Table 3 reports the estimation results of the dynamic copula with Markov-switching model. For all the geopolitical categories examined (namely US-China relations, global trade disputes, Gulf tensions, European fragmentation, North Korea conflict, Russia-NATO tensions and major terror attacks indexes), the parameters ω_H and ω_L of dynamic copula with Markov-switching reveal that the level of dependence between Bitcoin returns and geopolitical risk proxies (with the exception of Gulf tensions and Russia-NATO tensions indices) is positive and statistically significant in crisis period (ω_H), whereas it appears insignificant or relatively moderate in low volatility or normal states (ω_L). Likewise for the reaction of gold returns to all the geopolitical risk indices under study, except North Korea conflicts. Moreover, the significance of parameters φ and ψ indicate that rank correlations are time-varying. The probabilities π_{HH} and π_{LL} consistently indicate longer duration of the high dependence regime for the majority of cases.

Summing up, our findings reveal that both Bitcoin and gold can serve as safe havens against geopolitical risk. There exist, however, sharp heterogeneities across the different indicators of geopolitical risks. Such heterogeneity motivates us to develop a composite indicator, dubbed *GRCI*, by synthesizing the aforementioned risk indexes. To this end, to account for the elemental risk dynamics, and then allows detecting the core effects of geopolitical risk on Bitoin and gold returns, which are not specific to a particular measure of this phenomenon.

Table 4
CPA: The correlation matrix between the geopolitical risk indicators.

	UCR	*GTD*	*GT*	*EF*	*NKC*	*RNT*	*TA*
UCR	1	0.81345	0.51578	0.46781	0.30451	0.42567	0.09832
GTD		1	0.49632	0.55042	0.36821	0.59349	0.14789
GT			1	0.20342	0.24178	0.41942	0.07656
EF				1	0.21552	0.18764	0.42987
NKC					1	0.27692	0.04100
RNT						1	0.04318
TA							1

Notes: *UCR*: the US-China relations risks index; *GTD*: the global trade index; *GT*: the uncertainty over Gulf tensions; *EF*: the uncertainty related to the European fragmentation; *NKC*: the uncertainty over the North Korea conflict; *RNT*: The uncertainty over Russia-NATO tensions; *TA*: the uncertainty surrounding major terror attacks.

4.2. The construction of geopolitical risk composite indicator

Having determining dissimilar responses of Bitcoin and gold returns to the different geopolitical risk indicators considered throughout this study, we attempt in the following to identify their common determining factor which we interpret as the *GRCI*. For this purpose, we conduct a Principal Components Analysis (PCA). The latter aims at determining how various variables change in relation to each other, or how they are related. This is attained by transforming correlated original variables into a new set of uncorrelated time series using the correlation matrix. The new variables are linear combinations of the original ones and are sorted into descending order based on the amount of variance they explain of the original set of series. In other words, the main objective of the PCA investigation is to take Q variables to $x_1, x_2, \ldots x_Q$ and find linear combinations of these allowing to generate the principal components $Z_1, Z_2, \ldots Z_Q$ that are uncorrelated following:

$$Z_1 = a_{11}x_1 + a_{12}x_2 + \ldots + a_{1Q}x_Q \tag{6}$$

$$Z_2 = a_{21}x_1 + a_{22}x_2 + \ldots + a_{2Q}x_Q \tag{7}$$

$$Z_Q = a_{Q1}x_1 + a_{1Q2}x_2 + \ldots + a_{QQ}x_Q \tag{8}$$

The next step is to choose the principal components that preserve the greatest amount of the cumulative variance of the original data. Note that relatively moderate correlation in the principal components is a good feature as it implies that the principal components represent distinct statistical dimensions in the dataset.

The weights a_{ij} applied to the series x_j in Eqs. (6),(7) and (8) are selected so that the principal components Z_i satisfy two main conditions (i) they are uncorrelated, and (ii) the first principal component represents the maximum portion of the variance of the data, whereas the second principal component accounts for the maximum of the remaining variance.

4.3. Measuring the responses of Bitcoin and gold to the geopolitical risk composite index

The correlation matrix between the different geopolitical risk factors used is summarized in Table 4. The highest correlation is found between the sub-indicators *UCR* and a coefficient of 0.81, followed by the correlation among *RNT* and *GTD* with a coefficient of 0.59. The geopolitical risk indicators are likely to move together, implying that there is a common geopolitical risk component to all these proxies. We thereafter seek to identify this common component by developing a geopolitical risk composite indicator using the CPA.

From the above correlation analysis, we can also observe that the independent variables (*UCR, GTD, GT, EF, NKC, RNT* and *TA*) are heavily correlated with each other, hence we deduce the presence of multicollinearity in the data. There are several techniques to remove multicollinearity from the dataset. For instance, we can drop a few of the highly correlated hidden features, but that may lead in loss of information and is also a not effective tool for data with high dimensionality. The idea is to minimize the dimensionality of the data and preserve the maximum variance by employing the Principal Component Analysis (PCA) algorithm. The latter mainly consists of matrix factorization to lessen the dimensionality of data into lower space. To do so, we have to find the percentage of variance explained (%VE) as dimensionality collpases.

$$\%VE = \frac{1}{\sum_{i=1}^{d}\lambda_i} * \sum_{j=1}^{k}\lambda_j \tag{9}$$

Where λdenotes the eigenvalue; d refers to the number of dimensions of original datasets; $k$corresponds to the number of dimensions of new feature space.

From Fig. A2 (in Appendix), we find that the total variance of data captured by component 5 is about 0.97. Since 97 % of the cumulative variance is detected by the component 5, we consider only five components (in particular, the global trade (*GTD*), the US-China relations risks (*UCR*), the risks from the European fragmentation (*EF*) as well as market focus on risks such as Gulf tensions (*GT*) and the North Korea conflict (*NKC*)). By reducing the dimensiality of data using PCA, the variance is preserved by 97 % and the multicollinearity of data is removed.

Table 5
Estimation of the dynamic copula with Markov-switching: Bitcoin (vs. gold) and the composite geopolitical risk index derived from PCA.

	BTR-GRCI	*GR-GRCI*
ω_H	0.3145*** (0.0001)	0.2195** (0.0034)
ω_L	0.0562 (0.3084)	0.0013* (0.0156)
φ	0.2659* (0.0110)	0.3144*** (0.0007)
ψ	0.1955** (0.0062)	0.3628*** (0.0000)
π_{HH}	0.8672** (0.0019)	0.9652*** (0.0004)
π_{LL}	0.1546* (0.0112)	0.1876** (0.0019)

Notes: *BTR*: Bitcoin returns; *GR*: Gold returns, *GRCI:* the GPRs composite indicator synthesizing the core impacts of geopolitical risks; $\omega_H(\omega_L)$ correspond to the level of dependence between Bitcoin/gold returns and geopolitical risk proxies in crisis period (low volatility or normal states). φ and ψ refer to the rank correlations; Superscript *H (L)* indicates the high (low) dependence regime. π_{HH} (π_{LL}) is the probability of staying in the high (low) dependence regime. Notes: *, ** and***denote statistical significance at the 10 %, 5% and 1% levels, respectively.

Then, using the developed synthetic index, we can determine the core impacts of geopolitical risk on Bitcoin returns (and then, on gold returns). Table 5 reports the estimated parameters of the regression of *BTR* and *GR* on the of the geopolitical risk composite index. For investors wanting to use these safe haven features of Bitcoin and gold to protect against changes in geopolitical risks, it is very useful to discover whether the responses of Bitcoin and gold to GPRI is consistent across all states, or concentrated in particular times of geopolitical stress. This study assesses this property by accounting for periods of high and low geopolitical risk. We document that both Bitcoin and gold prices increase significantly amid heightened geopolitical risk (i.e., in high risk scenario). This deeply suggests that holding a diversified portfolio composed of Bitcoin or gold could help safeguarding against exposure to unforeseen geopolitical risks. By comparison with most other investment assets, gold and Bitcoin have the advantage of being free of counterparty risk: their perceived values are inherent, rather than relying on any other institution, such as a government, or central banks.

4.4. Discussion of results

Our findings reveal that both Bitcoin and gold can serve as safe havens against in times of rising uncertainty. From these results, the logical implication for investors and traders is that Bitcoin and gold could be useful assets to diversify away the price risk associated with sudden events or surprises. This outcome seems consistent with prior studies on the hedging and safe haven characteristics on Bitcoin and gold. For instance, Bouoiyour et al. (2017) found a positive linkage between Bitcoin and gold prices in periods of high economic uncertainty, but under bullish market regimes, this relationship appears much less pronounced. They attribute the positive correlation between Bitcoin and gold to the fact that both Bitcoin and gold represent investments that people seek because they are safe and risk minimal. Consistently with Bouri et al. (2020a), we show that the safe-haven properties of Bitcoin and gold may vary over different time-horizons or from one regime to another, with superiority of Bitcoin. Our results are also in line with Al-Yahyaee et al. (2019); Bouoiyour et al. (2019a); Bouri and Gupta (2019); Bouri et al. (2020b) and Su et al. (2020) providing evidence that Bitcoin acts as hedge in regard to growing geopolitical risks. Despite their distinct features, we consistently find that both assets may serve as a safe haven in times of distress but for different reasons, consistently with Bouri et al. (2020a); Będowska-Sójka and Kliber (2021) and Al-Khazali et al. (2018).

Gold is widely known for its ability to safeguard against unforeseen risks. In other words, it is universally accepted as the ideal hedge and safe haven asset, and central banks, governments and individual investors turn to it during periods of distress. This yellow metal has, therefore, a long history of being a reliable store of wealth. However, Bitcoin is a nascent market, less known as safe haven, and necessitates advanced knowledge in data processing. The Bitcoin market is also small, less structured and less regulated than the gold market. For gold, central banks, governments and individual investors would often perceive it as the ideal safe haven against uncertain exposure, which was traditionally its most common use. For Bitcoin, its limited supply and its growing popularity certainly raise its value. Moreover, the potential factors determining gold are known by market participants and academics, whereas the price formation of Bitcoin is very complex. Also, Bitcoin unlike gold suffers from information asymmetry, as its system is relatively complex and may not be effortlessly understood by all users.

Moreover, Bitcoin is highly volatile compared to gold. Looking at Fig. A3 (in Appendix), we observe the median daily change for each of Bitcoin and gold during a moving 200-day period, we clearly note that the gold's median change (the black line) since 2011 does not exceed 0.5 percent. This implies that on a majority of days, an investor holding gold should expect a profit or loss of around 0.5 percent. For the same time period, we notice that Bitcoin's median change (the red line) is approximately 1.7 percent, which indicates that an owner of Bitcoin should expect a gain or loss of about 1.7 percent each day, more than three times what a gold owner must bear. We observe also that Bitcoin's rolling 200-day median return is much more volatile than that of gold. In other words, the volatility gap between gold and Bitcoin seems larger. Therefore, one can expect that investors and traders who choose gold as a refuge for their investments are not the same as those who opt for Bitcoin. In fact, Bitcoin is generally chosen as an attractive investment. It is extremely volatile and its speculative behavior enables investors and traders to earn supernormal returns in a short-time span. It also provides better divisibility than gold. Nevertheless, the yellow metal has been viewed throughout civilization as an effective safe haven used to store value, and has never gone to zero in recorded history. Besides, gold as a physical asset is protected by a strong property law which is proven and universally understood. Furthermore, units of gold may be more interchangeable than Bitcoin. In fact, gold is gold anywhere in the world, but the transparency of the Bitcoin blockchain could enable merchants to discriminate against certain coins on the basis of their past owners and/or transaction history.

In brief, both Bitcoin and gold can be depicted as valuable assets. Bitcoin, nevertheless, has some advantages over gold including its fungibility, pre-determined supply, and its growing availability as a transaction tool. Also, its utility and value would rise over time as Bitcoin adoption continues to increase, and mining difficulty continues to grow. Bitcoin adoption will no doubt expand with increased institutional adoption. Thus, Bitcoin, although speculative and very volatile and its legal status remains undefined or changing in many countries at present, represents a great technological advance over gold and other physical commodities.[2]

5. Conclusions

The present research is the first, to the best our knowledge, to compare the safe haven abilities of Bitcoin and gold with respect different geopolitical risk indicators (namely US-China relations, global trade disputes, Gulf tensions, European fragmentation, North Korea conflict, Russia-NATO tensions and major terror attacks). To do so, we carry out a flexible copula approach to shed new light on Bitcoin and gold safe haven statuses under low and high geopolitical risk regimes. Having shown heterogeneous Bitcoin and gold reactions to the seven aforementioned geopolitical risk measures, we develop a composite index that summarizes the seven factors within one risk measure.

Our results consistently reveal a positive and significant relationship between Bitcoin returns and geopolitical risk under rising geopolitical risk periods; similarly for gold returns. The logical implication emerging from this assessment is that the crude oil could have a potential role to play in risk management. The novelty of this study lies on providing robust evidence that the abilities of Bitcoin and gold to act as a buffer against exposure to unforeseen geopolitical events depends on whether the risk is low or high and on the geopolitical risk measure used. We can explain the positive and significant responses of both Bitcoin and gold prices to the various geopolitical risk proxies by investors' high-risk aversion during rising uncertainty episodes and readiness to shift back to more beneficial assets throughout a recovery in financial markets. Both gold and Bitcoin tend to be resilient during market crashes because they are negatively dependent on risky assets. Although stock markets benefit from stability, gold and Bitcoin benefit largely from market volatility. If risky assets collapse, worries increase, and investors typically seek out the safe haven features of gold and Bitcoin. But investors should be mindful that Bitcoin is also linked to risks. Scandals and frauds are rampant in the Bitcoin ecosystem. Bitcoin is an entirely speculative asset, a characteristic that may be taken against Bitcoin, and as being a medium of exchange and store of value. In short, the excessive volatility, the speculation and the multiple unknowns involved in the development of this cryptocurrency forestalls it from being viewed as a well-established hedge, safe haven and a diversifier asset.

Given the fact that market participants have different expectations and anticipations, this work can be extended by relaying on the time investment horizons using the Maximal Overlap Discrete Wavelet Transform (MODWT). Such analysis may be highly relevant for mitigating exposure to rising geopolitical risks through portfolio construction and diversification, as time-varying responses to GPRs will be discovered. More particularly, insights regarding the variant relationship between Bitcoin and GPRs (in comparison to gold) can be considered as potential input to make the most appropriate hedging strategies as well as the optimal portfolio allocations. Another possible avenue of research is to determine the co-jumps between Bitcoin and gold prices and geopolitical risks. More research is needed to better understand the huge uncertainties surrounding further geopolitical tensions and the COVID-19 recovery and their implications for Bitcoin and gold markets and particularly how to efficaciously safeguard against heightened uncertainty.

CRediT authorship contribution statement

Refk Selmi: Writing – Original draft preparation, Conceptualization, Methodology, Estimation, Software, Data curation, The preparation of the revised version. **Mark Wohar**: Editing and Supervision (orginal draft and revision). **Jamal Bouoiyour**: Editing and Supervision (original draft).

Declaration of Competing Interest

None.

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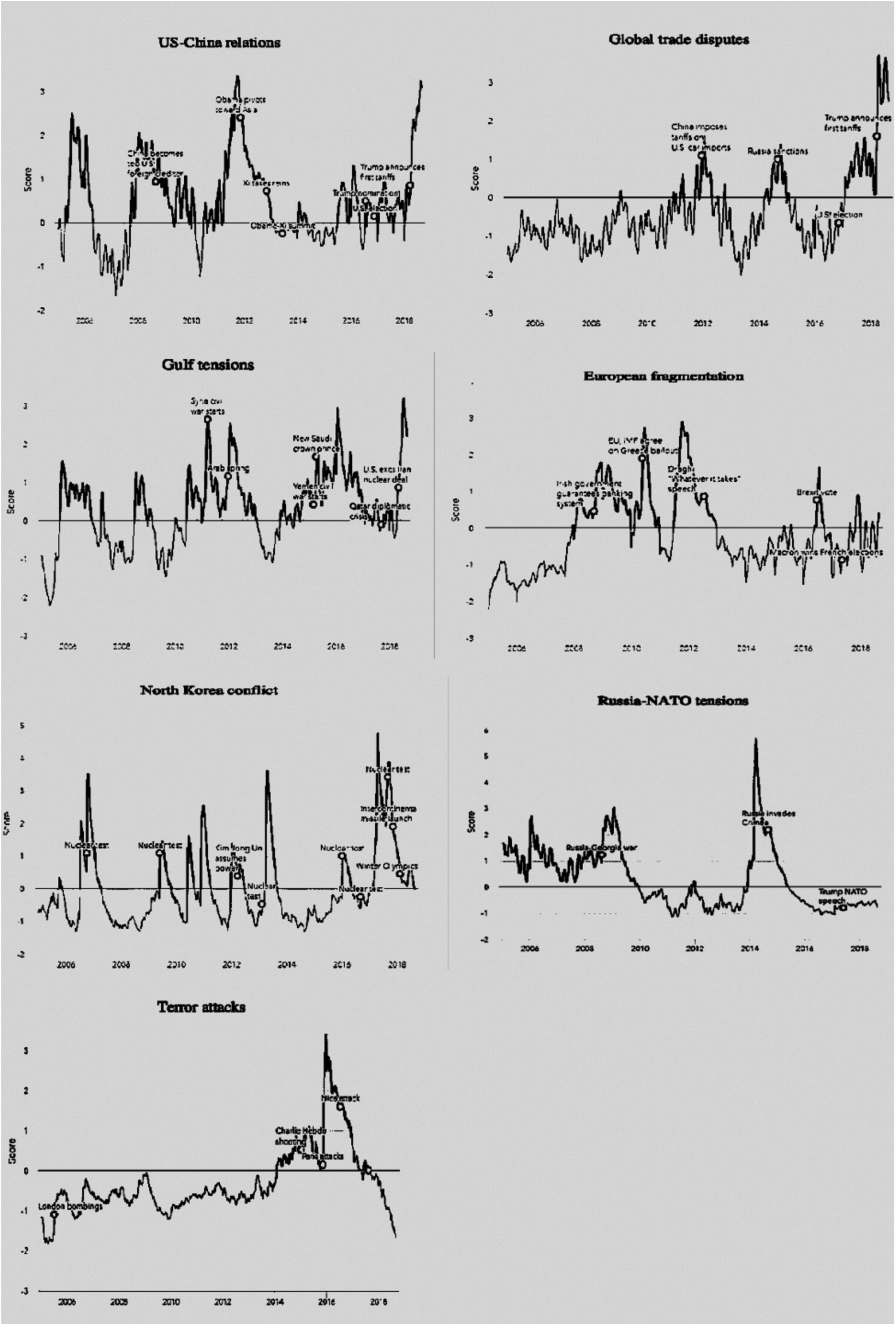
Appendix A

See Fig. A1

[2] Table A2 (Appendix) displays a summary of the differences and similarities between Bitcoin and gold.



(caption on next page)

Fig. A1. The different geopolitical risk indexes used throughout this study.

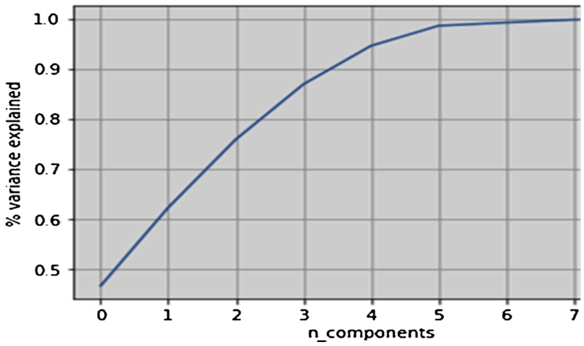


Fig. A2. The CPA: the percentage of variance explained versus the number of dimensions.

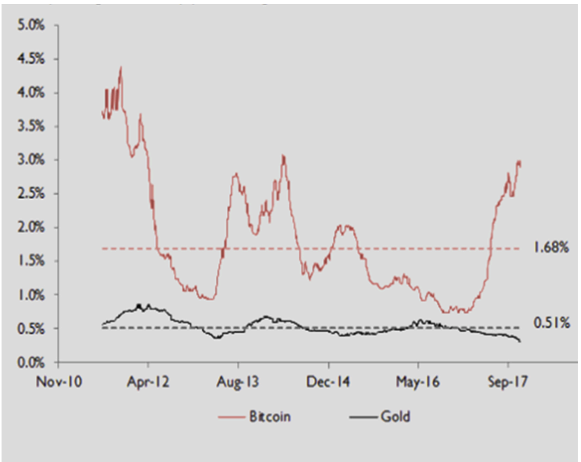


Fig. A3. 200-day rolling median daily price changes: Bitcoin vs. Gold.

Table A1
Data, definitions and sources.

	Variables	Definition	Links of data sources
The dependent variable:	*BTR*	The Coin Desk Bitcoin Price Index represents an average of Bitcoin prices across leading Bitcoin exchanges.	https://www.coindesk.com/price/bitcoin
	GR	The gold price is measured in USD per ounce.	World Gold Council https://www.gold.org/goldhub/data/gold-prices
The independent variable: The geopolitical risk indicators	*UCR*	US-China relations: The United States sets tariffs on a large range of Chinese goods.	BlackRock Investment Institute https://www.blackrock.com/co/recursos/herramientas/blackrock-geopolitical-risk-dashboard
	GTD	Global trade disputes: The U.S. escalates trade tensions with the aim of minimizing its trade deficit. Consequently, Market sentiment deteriorates amid increased worries of a global trade war.	BlackRock Investment Institute https://www.blackrock.com/co/recursos/herramientas/blackrock-geopolitical-risk-dashboard
	GT	Gulf tensions: The U.S. government has taken a rising confrontational stance in the Middle East strengthened by pulling the United States out of the Iran nuclear deal.	BlackRock Investment Institute https://www.blackrock.com/co/recursos/herramientas/blackrock-geopolitical-risk-dashboard
	EF	The European fragmentation: The European debt crisis, the Brexit vote and the rise of populism in Europe seen by the populist	BlackRock Investment Institute

(*continued on next page*)

Table A1 (*continued*)

Variables	Definition	Links of data sources
	orientation of Italy's new government seems to stir disputes with European partners, slowing eurozone integration.	https://www.blackrock.com/co/recursos/herramientas/blackrock-geopolitical-risk-dashboard
NKC	The risk of military North Korea conflict.	BlackRock Investment Institute https://www.blackrock.com/co/recursos/herramientas/blackrock-geopolitical-risk-dashboard
RNT	Russia-NATO tensions : Russia and the West have attained an uncomfortable situation (i.e., mutual recriminations and military preparations)	BlackRock Investment Institute https://www.blackrock.com/co/recursos/herramientas/blackrock-geopolitical-risk-dashboard
TA	Terrorist attacks: Terror attacks in major Western European cities prompting wider casualties.	BlackRock Investment Institute https://www.blackrock.com/co/recursos/herramientas/blackrock-geopolitical-risk-dashboard

Table A2
A summary of dis/similarities between Bitcoin and gold.

Characteristics	Bitcoin	Gold
Fungible (i.e., interchangeable)	High	High
Portable	High	Moderate
Verifiable	High	Moderate
Established history	Low	High
Divisible	High	Moderate
Secure (i.e., cannot be conterfeited)	High (cryptographic)	Moderate (physical)
Volatile	High	High (but less than Bitcoin)
Speculative	High	Moderate
Easily transactible	High	Low
Scarce (predictable supply)	Fixed-21 million bitcoin	High
Sovereign (Government issued)	Low	Low
Decentralized	High	Low
Smart (programmable)	High	Low

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